

Power equations



1

a $x = 3$

b $x = 8$

c $x = \frac{8}{5}$

d $x = 6$

e $x = 2$

f $x = 6$

g $x = 5$

h $x = 2$

Exponential equations

2

a $x = 8$

b $x = \frac{2}{9}$

c $x = 3$

d $x = 0$

e $x = -2$

f $y = -3$

g $x = -3$

h $x = \frac{1}{3}$

i $n = \frac{1}{3}$

j $x = -2$

3

a i $2^{3x} = 2^2$

ii $x = \frac{2}{3}$

b i $2^{4x} = 2^{-1}$

ii $x = -\frac{1}{4}$

c i $2^{-10} = 2^{2x}$

ii $x = -5$

d i $2^{\frac{x}{2}} = 2$

ii $x = 2$

4

a $y = 4$

b $k = -2$

c $y = \frac{3}{2}$

d $x = 2$

e $x = 2$

f $x = \frac{7}{3}$

g $x = -7$

h $x = \frac{21}{19}$

i $y = \frac{11}{6}$

j $x = -\frac{41}{6}$

k $x = 5$

l $x = 1$

m $x = 1 + \frac{2k}{3}$

n $x = 5$

o $x = \frac{5}{2}$

p $x = 4, x = -1$

q $x = 3$

r $x = \frac{4}{1+n}$

5

a i $m = 2^x$

ii $x = 3, 0$

b i $m = 2^x$

ii $x = 3, 2$

c i $m = 2^x$

ii $x = -1, 3$

d i $m = 4^x$

ii $x = 0, 3$

e i $m = 2^x$

ii $x = 0, 2$

f i $m = 3^x$

ii $x = 1, 2$



Exponential equations and logarithms

6

a $3 < x < 4$

c $-4 < x < -3$

b $3 < x < 4$

d There is no real solution.

7

a i $x = \frac{\log 5}{\log 13}$ ii 0.627

b i $x = \frac{\log \frac{1}{11}}{\log 5}$ ii -1.490

c i $x = \frac{\log 2}{\log 3}$ ii 0.631

d i $x = \frac{\log 6.4}{\log 4}$ ii 1.339

e i $x = \frac{\log 5}{\log 0.4}$ ii -1.756

f i $x = \frac{\log 3125}{\log 5}$ ii 5

g i $x = \frac{\log 6}{\log \frac{1}{2}}$ ii -2.585

h i $x = \frac{\log 19\,207}{\log 27}$ ii 2.993

8

a $x = \frac{\frac{\log 70}{\log 4} + 8}{2}$

b 5.532

9

a i Step 2, when bringing the a out of the index it should multiply the logarithm and not add to it.

ii $a = \frac{\log 40}{\log 9}$

iii 1.679

b i Step 3. The 2 and 89 are in the incorrect positions in the logarithm.

ii $a = \frac{\log 89}{\log 2}$

iii 6.476



Applications

10

a $t = 5$ hours

b $t = 10$ hours

c $t = 11.55$ hours

11

a 500 mice

b 20%

c 6.03 months

12

a 800 wallabies

b 15%

c 8.53 years

13 6.180 hours